

Introductory Project (1DT308)

TIG Stack (Telegraf, InfluxDB, Grafana)

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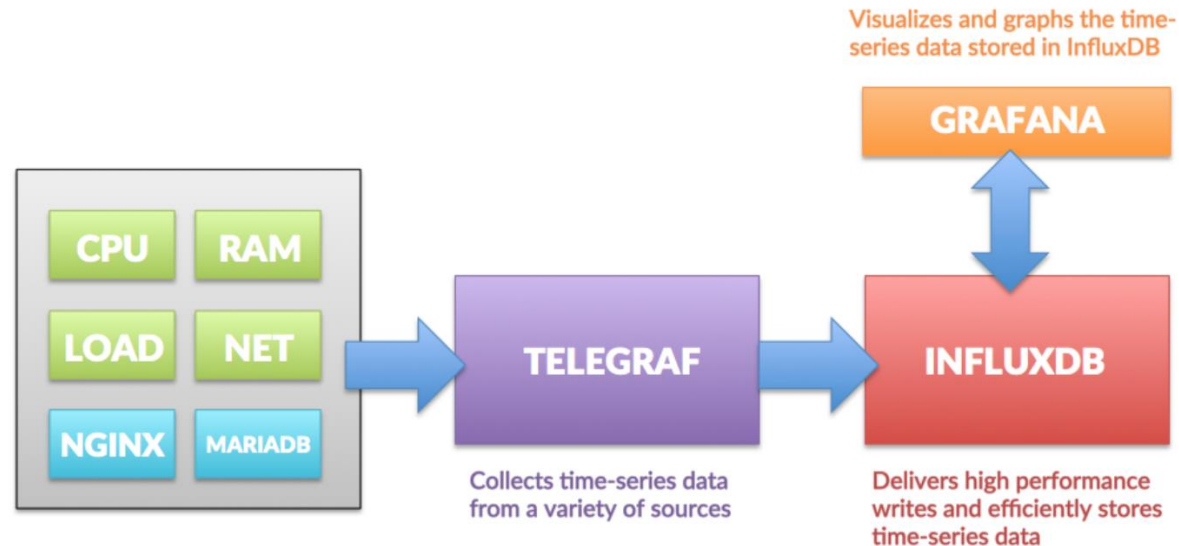


Introduction to TIG Stack (Telegraf, InfluxDB, Grafana)

Telegraf: Data collection agent that pulls data from various sources (e.g., MQTT) and sends it to InfluxDB.

InfluxDB: Time-series database used to store sensor data.

Grafana: Visualization tool to create dashboards from the data stored in InfluxDB.



What is Mosquitto?

Mosquitto:

- An open-source MQTT broker that can be run locally on your machine, offering control over data and security.

Adafruit IO:

- A cloud-based MQTT platform.
- Ideal for users who prefer not to manage their own MQTT broker and infrastructure.



Mosquitto and Adafruit IO Comparison

Feature	Mosquitto	Adafruit IO
Deployment	Local (on personal machines)	Cloud-based
Cost	Free and open-source	Free tier, paid plans for advanced features
Customization	Fully customizable, open-source	Limited customization
Data Privacy	Local storage, full control	Cloud storage (Adafruit's servers)
Ease of Use	Requires setup and management	Easy setup with user-friendly interface
Scalability	Can scale from small to large deployments	Limited scaling, depends on the plan
Advanced Features	Full MQTT support (QoS, retention, etc.)	Basic MQTT features
Ideal For	Private IoT projects, large-scale systems	Rapid prototyping, small projects



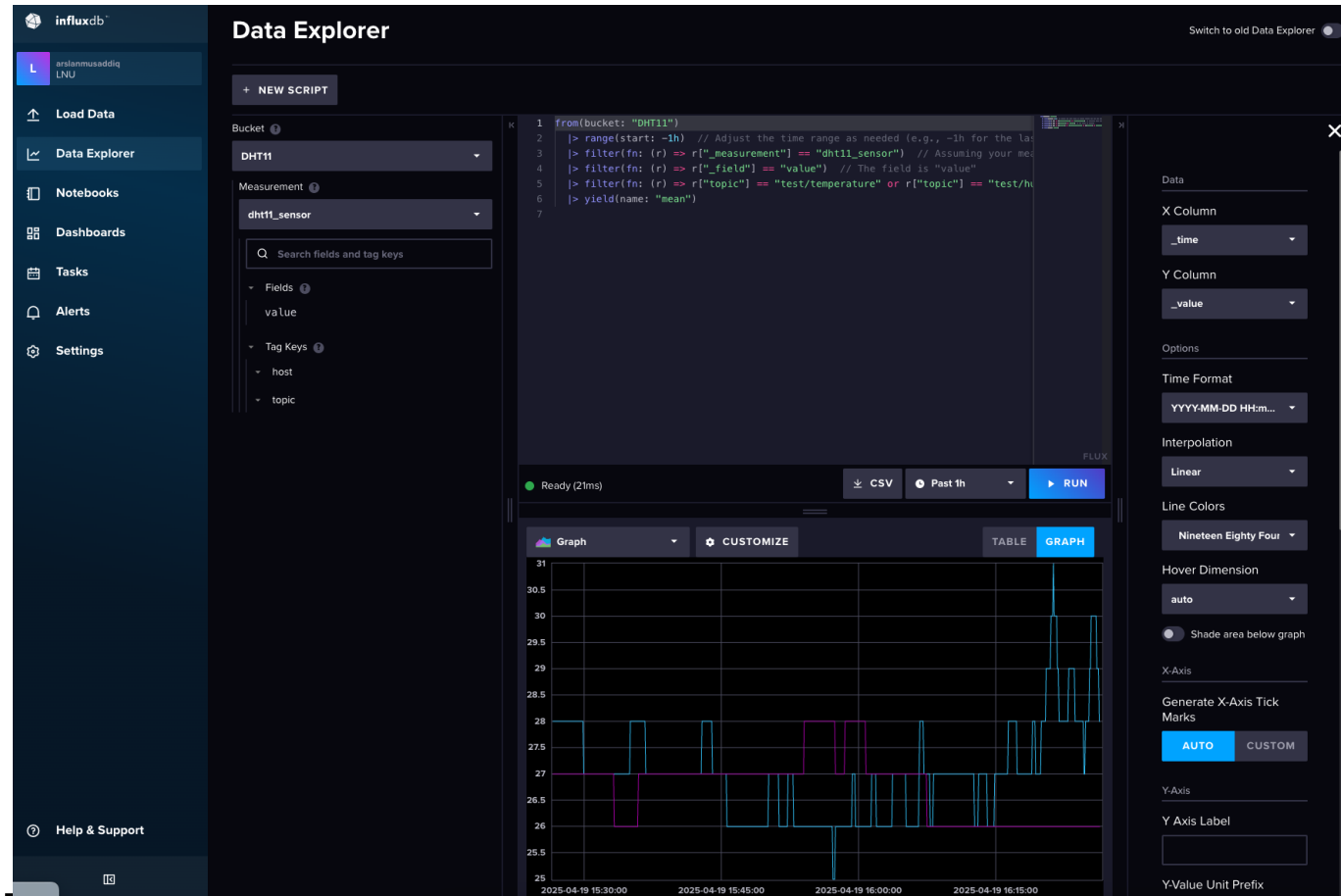
What is Telegraf?

- **Telegraf:** An open-source agent used for collecting and sending metrics to databases.
- **Role in the Pipeline:** Collects MQTT data (e.g., temperature and humidity from the sensor) and forwards it to InfluxDB



What is InfluxDB?

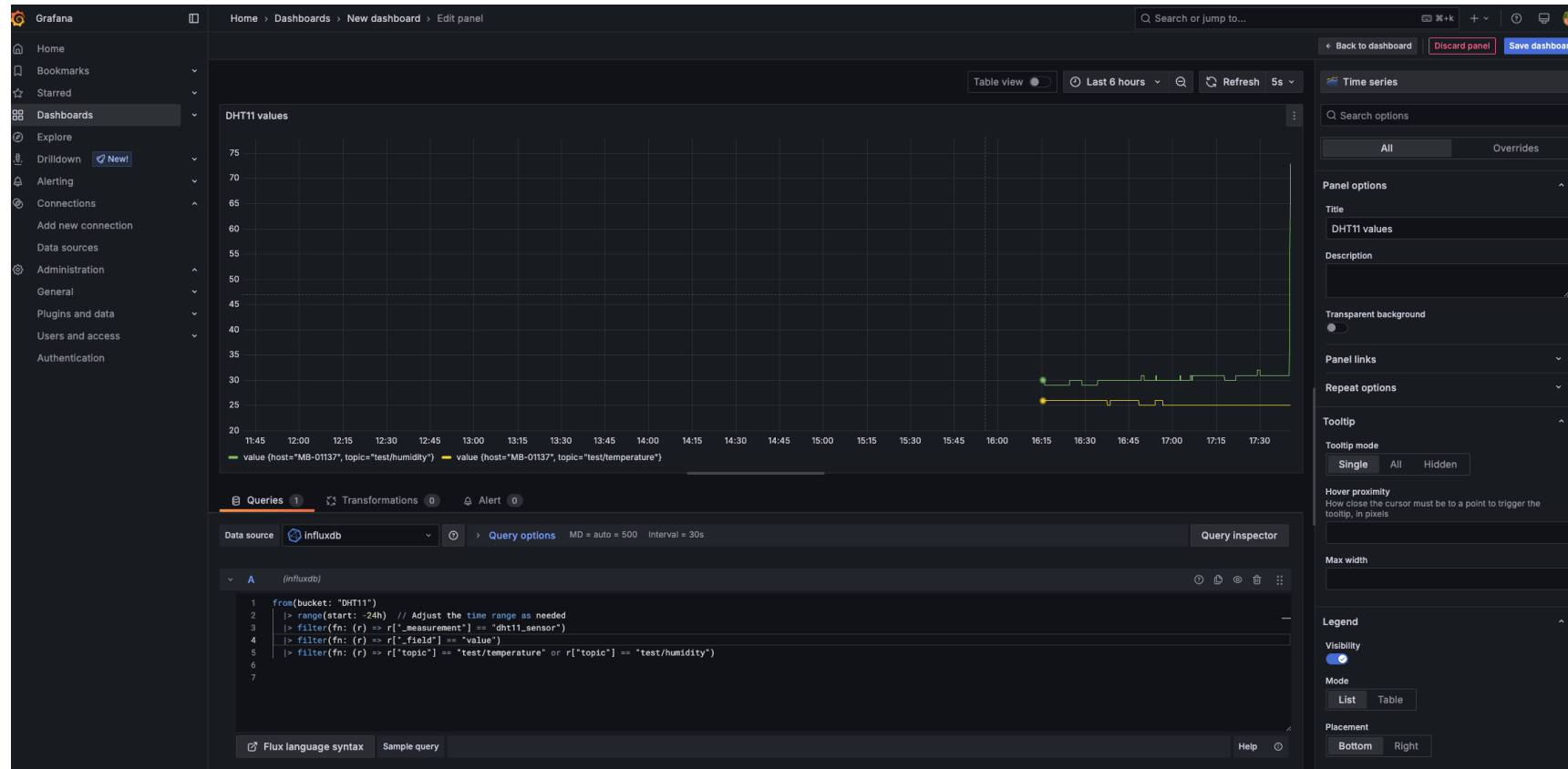
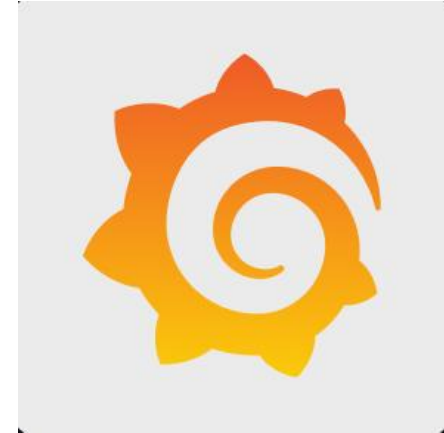
- **InfluxDB** is a time-series database optimized for storing, querying, and analyzing time-stamped data.
- **Query Language:** InfluxQL (SQL-like syntax) or Flux (more powerful query language).



What is Grafana?

Grafana: A powerful open-source platform for visualizing time-series data.

Role: Pulls data from InfluxDB and creates interactive dashboards.



How Mosquitto Integrates with the TIG Stack?

- Telegraf can subscribe to topics on Mosquitto to collect data in real-time.
- Mosquitto serves as the messaging backbone that allows devices (e.g., IoT sensors) to send data to Telegraf.
- InfluxDB stores the time-series data collected via Telegraf.
- Grafana visualizes the data, including real-time data from IoT sensors transmitted via Mosquitto.



Setting Up the TIG Stack with Mosquitto (Overview)

- **Step 1:** Install and configure **Mosquitto** as the MQTT broker.
- **Step 2:** Install and configure **Telegraf** to collect data from Mosquitto topics.
- **Step 3:** Set up **InfluxDB** to store the time-series data.
- **Step 4:** Use **Grafana** to visualize the collected data on interactive dashboards.

Go to the following link (Lab 2) for complete and detailed instructions:

Lab 2 - Visualizing IoT Sensor Data using TIG Stack

https://gitlab.lnu.se/1dt308/content/-/blob/master/labs/2_TIG%20Stack.md?ref_type=heads

Take your time going through it, and feel free to ask if anything's unclear.





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